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A foundation model for nucleotide sequences

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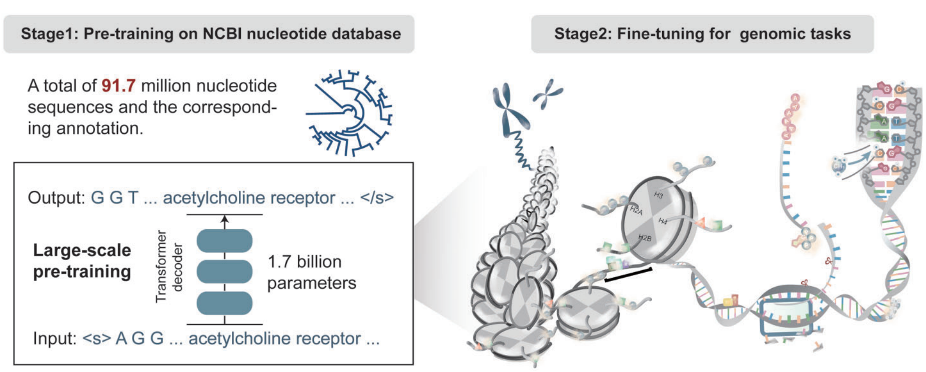
Abstract

Foundation models have demonstrated exceptional performance across diverse downstream tasks. However, in genomics and transcriptomics, the integration of nucleotide sequences with their rich annotations remains underexplored, potentially limiting model generalizability across species and biological contexts. Here, we introduce OmniNA (Omni-applicable foundation model for nucleic acid), a self-supervised generative foundation model trained on 91.7 million nucleotide sequences and their associated annotations, totaling 1076.2 billion bases and 197 million words spanning diverse species. Unlike most existing approaches that focus solely on functional genomic element recognition, OmniNA jointly learns from both sequences and annotations, leveraging their complementarity to enhance semantic understanding and representation learning. We demonstrate that OmniNA captures sequence grammar and annotation semantics, facilitating robust transfer across nucleotide-level tasks. OmniNA can be fine-tuned under natural language paradigms and achieves state-of-the-art or competitive performance in 23 benchmarks, including sequence detection and species classification. Its learned representations also help reveal mutation effects on DNA and RNA processing. We release the model publicly as a community resource for genomics and transcriptomics research. OmniNA represents a step toward advancing foundational modeling by integrating annotation-aware learning, offering a powerful tool for genomics and transcriptomics research.

Graphical abstract



Introduction

Artificial intelligence is currently undergoing a paradigm shift toward foundation model that leverages large language model (LLM) incorporating millions of parameters trained on extensive data, making them versatile for a wide range of predictive applications [1]. These models learn underlying data

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patterns and structures from massive unlabeled data, facilitating innovative applications through transfer learning [2–4]. This paradigm has emerged as a transformative technology with many successful examples in various fields, such as the generative pre-trained transformer [5] in natural language processing and Stable Diffusion [6] in vision-language. In the ever-evolving landscape of genomics, the advent of foundation model has emerged as a transformative force, propelling our capacity to decode the intricate language embedded within nucleic acid sequences. While early efforts have primarily focused on modeling raw nucleotide sequences, biological annotations—ranging from functional elements to regulatory labels—carry rich semantic context that is often overlooked. These annotations serve as the "semantic labels" of the nucleotide language, offering crucial cues that complement the syntactic patterns encoded in the sequences themselves. Against this backdrop, we present a large-scale generative pre-trained foundation model specifically tailored for nucleic acid sequences across diverse species and biological contexts. By jointly modeling both sequences and their associated annotations, our approach seeks to more fully capture the compositional grammar and semantic structure underlying genomic information, enabling robust generalization and interpretability across tasks.

The rapid progress of deep learning methodologies in genomics has been propelled by the escalating volume and diversity of genomic data. Existing models, such as Enformer and TIGER [7, 8], have made contributions to specific genome tasks. Enformer leverages a transformer architecture [9] for expression and chromatin state prediction, while TIGER is a deep convolutional network predicting off-target activity of RNA-targeting CRISPR guide RNA sequences [8]. However, these deep learning approaches are tailored to specific genome tasks, limiting their generalization ability to other tasks. Additionally, as the model parameters are trained from scratch with labeled data, they cannot be transferred across tasks. In the ever-evolving landscape of genomic data, it is crucial to develop foundation models capable of adapting to various nucleotide sequences and transferring knowledge to a broad spectrum of genomic tasks.

This study introduces OmniNA (Omni-applicable foundation model for nucleic acid), a foundation model for nucleotide sequences. OmniNA was pre-trained on a scale of 91.7 million (M) nucleotide sequences encompassing 1076.2 billion (B) bases across a global species and biological context. Unlike previous works, our approach integrates nucleotide sequence and corresponding text annotations into the learning process. To our knowledge, this is the first study to learn the largest publicly available nucleotide sequences and the annotations spanning a diverse array of species. This integration allowed the model to understand the semantics of biological annotations, fostering a more comprehensive representation, empowering it to perform a wide range of downstream tasks. Our analysis demonstrates that the model learned the syntax of nucleotide sequences by analyzing the learned representations of sequences. Subsequently, a comprehensive evaluation of our pre-trained model was conducted to assess its ability to adapt to new captions and handle multi-tasks in one model. After fine-tuning, the model gains the capability to unveil the mutational effects through its comprehension of nucleotide grammars.

Materials and methods

Data processing for OmniNA pre-training

We curated sequence data along with corresponding textual annotations from the Nucleotide (NT) Database of the National Center for Biotechnology Information (NCBI) up to April, 2023. The NT database is composed of nucleotide sequences in FASTA format, accompanied by text annotations including organism names, taxonomic classifications, gene and gene product names, and functional descriptions. These annotations provide context for training, linking the sequences to relevant biological information. The annotations obtained from the NCBI NT database comply with the standards established by the International Nucleotide Sequence Database Collaboration [10]. This framework ensures consistency through controlled vocabularies and a standardized taxonomic hierarchy integrated with the NCBI Taxonomy database. The data encompass genomic DNA and RNA sequences derived from a wide range of species, enabling a diverse foundation for self-supervised learning tasks. To characterize the annotation landscape used during pre-training, we quantified the distribution of sequence types, taxonomic labels, and token embeddings derived from the curated annotations. The composition of sequence types and taxonomic ranks across the corpus is summarized in Supplementary Table S1 and visualized in Supplementary Fig. S1A–C, providing an overview of the species diversity represented in the dataset. We further parsed all textual annotations into 20 token categories, forming a structured label space for model training. These token embeddings were analyzed using UMAP, revealing coherent clustering patterns that reflect shared biological or functional themes (Supplementary Fig. S1D). A detailed description of each token cluster, including representative themes and example annotations, is provided in Supplementary Table S2. Employing a sliding window approach, sequences longer than 3000 bp were truncated to 3000 bp, while sequences with a length <200 bp were excluded. We filtered out sequences containing two or more consecutive N nucleotides. Subsequently, a total of 91 732 311 sequences were included, collectively comprising 1 076 183 408 570 bases. The textual annotations totaled 197 089 040 words. We randomly selected 98% of the data for training and allocated 1% each for validation and testing. We designed prompts to organize the sequence data with textual annotations as feature-ordered input for model pre-training. As shown in Supplementary Table S3, the prompt is composed of three modules: (i) an instruction, (ii) a nucleotide sequence, and (iii) an annotation.

Model architecture and pre-training

We established the OmniNA model with the LLaMA architecture [11], which is an autoregressive decoder-only transformer that adopts the self-attention mechanism. The model was first pre-trained in a self-supervised manner using nucleotide sequences and their corresponding text annotations from the NT database. The pretraining objective was next-token prediction, where the model learned to predict the subsequent nucleotide or annotation token given the preceding context. The data were tokenized using the byte pair encoding algorithm [12]. We trained the tokenizer using nucleotide sequences and text from the wiki-103 dataset [13]. SentencePiece [14] was employed as the subword tokenizer, resulting in a dictionary containing 32 001 tokens. Subsequently, the nucleotide sequences and text annotations were partitioned into sentence-based chunks. We included a special token < s > at the beginning and a special token </s > at the end of the sequence. Each chunk was further tokenized using the trained tokenizer. The maximum length of the input sequence was set to 601 tokens. The token embedding layer embedded the input tokens into embedding vectors. Rotary positional encodings [15] were added to the token embeddings. The transformer consists of multiple layers, each containing a multihead self-attention layer and a feedforward layer. We constructed three models with varying heads and layers, ranging from 66 M up to 1.7 B parameters (Supplementary Fig. S2). The transformer layer captures token–token interaction information using a multihead self-attention mechanism [9]. Given an input, the self-attention mechanism assigns an attention weight $\alpha_{i,j} > 0$ to each token pair *i,j*, where $\sum_j \alpha_{i,j} = 1$. Attention in the transformer is bidirectional. Attention weights $\alpha_{i,j}$ are computed by the scaled dot-product of the query vector of $i(Q)$ and the key vector of $j(K)$, followed by a softmax operation. Then, attention weights are used to produce a weighted sum of value vectors (V):

$$\text{Attention}\,(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

where d_k is the dimension of K. Before feeding into transformer layers, we normalize the input with root mean square layer normalization [16]. We replace the ReLU non-linearity of the traditional transformer architecture by the SwiGLU activation function [17]. The output of the last transformer layer is fed into a linear layer for autoregressive learning [5]. The sequence and annotations were trained in an autoregression. Specifically, the prediction at each time step depends on the preceding observations. For example, the prediction for the base b_t at time t is based on the preceding observation b_{t-1} at time $t - 1$:

$$b_t = \beta_0 + \beta_1 b_{t-1} + \beta_2,$$

where β_0, β_1, and β_2 are trainable parameters governing the autoregressive prediction. Mean squared error loss function was applied for autoregression learning, defined as:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(b_t - \hat{b}_t\right)^2,$$

where N represents the number of tokens in one input. A detailed model architecture is shown in Supplementary Fig. S3.

The model was trained using the AdamW optimizer [18], with the following hyper-parameters: $\beta_1 = 0.9$ and $\beta_2 = 0.999$. We use a learning rate of 1e−4 and a batch size of 2048 for 200 000 steps (Supplementary Fig. S2). The learning rate was decayed with a cosine schedule and a warm-up ratio of 0.03. We used a weight decay of 0.0001 and gradient clipping of 1.0. All experiments were run in Python v.3.8. Detailed software versions are PyTorch v.1.12.1, CUDA v.11.7, CUDNN v.8.5.0, transformers v.4.28, datasets v.2.11, and tokenizers v.0.13. The model was run on an NVIDIA DGX A100 system with 8 GPUs, each with 80 GB of memory.

Evaluation of OmniNA's generation ability

We primarily focus on the fidelity, diversity, reproducibility, and stability of the generations produced by the pre-trained OmniNA. Genomic contents are generated using beam search, unless otherwise specified. The model fidelity is initially assessed by examining the consistency between OmniNA-generated annotations and the ground truth annotations. Three annotation tasks, encompassing species, genomic types, and enzyme annotations, are employed for evaluation. We compiled a list of 172 species that appeared in the NCBI NT database for >1000 times. Subsequently, 100 sequences were randomly sampled for each of these 172 species. For the genome type annotation task, 100 sequences of coding sequence, intron, genome coding region, messenger RNA, small nucleolar RNA, long non-coding RNA, microRNA, ribosomal RNA, non-coding RNA, and transfer RNA were randomly sampled from the NCBI NT database. Informal names and abbreviations of taxonomies were collected from the GenBank taxonomy database (https://www.ncbi.nlm.nih.gov/taxonomy). In the enzyme annotation task, we collected 104 enzymes that appeared in the NCBI NT database for >1000 times. Following this, 100 sequences were randomly sampled for each of the 104 enzymes from the NCBI NT database. Additionally, we evaluated the fidelity of the sequence generation as the similarity between the generated sequence and the ground truth sequence. The similarity was evaluated as the cosine similarity between the token embedding of the sequences. The bacteria and virus coding sequences were collected from the NCBI NT database.

We further quantified the conservation of protein coding sequences as the mean cosine similarity among the sequences' token embeddings of the respective protein among different taxonomies. The cosine similarity ranges from −1 to 1, where higher values suggest greater similarity. The diversity of tokens was quantified with Shannon entropy from 10 000 randomly sampled sequences from the NCBI NT database. We consider exact matches as correct after applying a vocabulary mapping that converts model-predicted vocabulary to informal names and abbreviations. The Shannon entropy score ranges from 0 to a maximum value, where higher values suggest greater diversity and unpredictability of the tokens in a given set. The official gene names are collected from the NCBI database (https://ftp.ncbi.nih.gov/gene/). The reproducibility of model-generated text annotation is quantified as the similarity among the different repeats of the model generations. We randomly select 10 000 sequences from the NCBI NT library and generate annotations for 10 repeats using the nucleus sampling generation algorithm [19] with a $top\_p = 0.92$. The reproducibility is quantified as the mean cosine similarity among the token embeddings of the 10 repeats. We next evaluated the stability of the model-generated annotations, which was assessed as the consistency of the generated annotations among different truncations of the same sequence. We randomly sampled 10 000 sequences with lengths >1000 from the NCBI NT database and randomly truncated them into 10 fragments per sequence. The stability is quantified as the mean cosine similarity among the feature representations of the annotations.

Feature representation evaluation of the OmniNA model

For each token presented to OmniNA-1.7B, the model embeds it into a 2048-dimensional vector that encodes the characteristics of the token to the sequence context. We extracted the embedding vector of the vocab and nucleotide sequence from the token embedding layers of the pre-trained OmniNA-1.7B model. For each token, the average embedding vectors of the sub-tokens for each vocab or sequence are visualized with t-SNE dimensionality reduction. For each transformer layer, we further extract the feature representations of the nucleotide sequence as the output of transformer layers. The adjusted rand index (ARI) score is applied to evaluate the transformer layers in terms of sequence representation ability. For comparison, we applied the same procedure to extract embedding and transformer layer feature representations from DNABERT2 [20], Nucleotide Transformer-v2 [21], and Hyena-DNA [22]. When input sequences exceeded a model's maximum length, we truncated them symmetrically from the center to meet the input size constraints.

The ARI score measures the percentage of matches between the sequence clusters and labels. The resulting score ranges from −1 to 1, where a higher score denotes that the data point fits well in the current cluster. We used the Louvain community detection algorithm implemented in "tl.louvain" of the Scanpy package (version 1.7.0) for sequence clustering. In our study, the Louvain algorithm would generate many more sequence clusters than real sequence types when the resolution was 1 and far fewer when the resolution was 0.0001. Thus, we set the resolution parameter range from 0.0001 to 1 with a step of 0.001 and computed ARI score with the sklearn package (version 1.2.1) for each step. The maximum ARI score is employed as the final evaluation index.

Three specific tasks in terms of sequence conservation delineation, species differentiation, and functional sequence differentiation are applied for test. The sequence conservation of human genome is calculated with the GERP score [23]. Specifically, employing a sliding window approach, we truncated the genomes into non-overlapped sequences of 1000 bp and calculated the mean GERP score for each sequence. Sequences with the highest and lowest quartiles of GERP scores are defined as high- and low-conserved sequences, respectively. Bacteria and archaea 16S RNAs and fungi ITS regions were collected from the NCBI RefSeq database (https://www.ncbi.nlm.nih.gov/refseq/targetedloci/). The nucleotide sequences of SARS-CoV-2 were collected from GISAID (https://nextstrain.org/ncov/gisaid/global/6m). The coding sequences of functional proteins of bacteria and viruses were extracted from the NCBI NT database. The promoter sequences are collected as the 2000 bp around the transcription start site of the human GRCh38 reference genome. The human silencer and enhancer sequences are collected from the SilencerDB and VISTA enhancer browser [24–26]. The 5′ and 3′ UTR sequences are collected from human GRCh38 reference genomes. Transcription factor binding sequences are randomly sampled from TF ChIP-seqs of the encyclopedia of DNA elements (ENCODE) database [27]. CpG islands were sourced from the UCSC Genome Browser (https://genome.ucsc.edu/) and categorized into three bins based on the percentage of CpG content: <15%, 15%–30%, and >30%. The non-B structures were sourced from the following databases: G4Bank (G4), RloopDB (R-loop), IMSeeker (i-motif), InvertiaDB (IR), and NonB DB (α-phased, Z-DNA, short tandem repeats, and mirror repeats). The RNA sites were sourced from the following databases: ENCODE database (RBP), RMBase v3.0 (m6A/m1A modification data), IRESbase (IRES), AREsite2 (ARE), and miRDB (miRNA binding sites). For each of these features, we randomly sampled 5000 sequences from the relevant database, using them as labeled data.

Description of the downstream tasks

The model underwent fine-tuning for 23 tasks in a multitask framework. Supplementary Table S4 provides an overview of data resources and the splitting for training, validation, and test sets. During deployment, the model operates on sequence-only inputs; annotation tokens are neither required nor used. A detailed schematic illustrating the sequence-only inference pathway is provided in Supplementary Fig. S4. Below, we briefly outline these tasks. We focused on tasks related to human DNA elements, encompassing promoter, enhancer, silencer, insulator, and active promoter and enhancer identification. Promoter sequences, defined as 2000 bp around the transcription start site, were collected from GRCh38 and mm10 reference genomes. Enhancer, silencer, and CTCF sequences were sourced from VISTA enhancer browser, SilencerDB, and CTCFBSDB, respectively. Active promoter data were collected from MPRA-DragoNN. Active enhancer data were collected from DeepSTARR [28]. The data for PAS signal, TIS signal, splice site, and CRISPR off-target prediction were collected from previous studies [8, 29, 30]. The genome-wide methylome profiles of 39 human cell types were sourced from previous research [31]. For chromatin accessibility, histone occupation, and transcription factor (TF) binding site prediction, we downloaded the ATAC-seq, DNase-seq, and ChIP-seq in .bed format from the ENCODE database (Supplementary Table S5). We retained data meeting two criteria: (i) data with the highest audit tiers defined in the ENCODE database or two biological replicates and (ii) data with >1000 positive peaks. A total of 114 chromatin accessibility profiles from 89 cell lines were collected for chromatin accessibility prediction. A total of 884 TF ChIP-seq samples targeting 271 TFs across 56 cell lines were collected for TF binding prediction. A total of 248 histone ChIP-seq samples targeting 22 regions across 70 cell lines were collected for histone binding prediction. We defined a region with coverage RGB score >50 as a positive peak. We further filtered out peaks <200 bp or longer than 3000 bp. To balance data, we randomly sampled 6000 and 1000 peaks if the positive peak count overmuch for training and test, respectively. We further collected single-nucleotide substitutions, insertions, deletions, and synonymous variants from the ClinVar database [32] and the 1000 genomes dataset [33]. We then fine-tuned the model on a per-gene basis and selected genes with at least 100 pathogenic and 100 benign variants in the test set after deduplication. For benchmarking against EVE, we further restricted the evaluation to the intersecting gene set between both models, yielding 48 genes used for comparative analysis [34]. For each SNV, we collected sequences with a length 1000 bp centered on the mutation site for pathogenic or benign classification. The data for the pathogenic virus prediction task are collected from DeePaC [34]. The data for taxonomy prediction task are collected from DeepMicrobes [35]. For every task, we use mutually exclusive train/validation/test partitions; test sequences are never used for model training. We used original data splits from prior studies where available (eg. DeepSTARR, SpliceRover, and DeepGSR). ENCODE data were split into training, validation, and test sets (95:2.5:2.5) without overlap. SNP data were split per gene (80:1:19) to account for sample imbalance. See Supplementary Table S4 for details. Antibiotic-resistance genes (ARGs) and mechanism labels were obtained from CARD database [36]; we retained mechanism categories with ≥2000 sequences, yielding five high-prevalence mechanisms. Virulence genes (VGs) and functional types were obtained from VFDB [37]; for subtyping, we retained types with ≥100 sequences, yielding

11 categories. Negative examples were randomly sampled from protein-coding genes in NCBI RefSeq bacterial genomes and filtered by BLAST against CARD/VFDB to remove significantly high-similarity hits. Phage–host labels of bacteriophage were collected from PhageScope [38]. We retained 12 host taxa with ≥1000 sequences each. Host labels of eukaryotic viruses were collected from DeepHost dataset [39]. We retained 144 host taxa with ≥50 sequences each. For both bacterial functional sequence and host specificity prediction tasks, the data were split into training, validation, and test sets in an 8:1:1 ratio. cell-free RNA sequences of pancreas patients and normal persons were collected from GSE136651. After adapter trimming, we randomly sample 30-bp fragments and concatenate 70 fragments to form one input sequence per patient; 1000 inputs per patient are generated. Model logits are averaged across a patient's inputs (threshold 0.5) to obtain the patient-level label. We conduct five-fold cross-validation with an 80/20 split stratified by patient, ensuring no patient overlap between training and test folds.

Apply the model for multi-task fine-tuning

In the fine-tuning phase, the model was further trained using supervised learning on task-specific datasets. Here, the model was trained to generate text annotations (e.g. biological responses or answers to questions) given nucleotide sequences as input. We adopted an autoregressive learning framework, where the model predicted the next token, with the targets being annotation tokens rather than nucleotide bases. As shown in Supplementary Table S3, each task instance includes a prompt composed of four components: (i) an instruction, (ii) a nucleotide sequence, (iii) a test question, and (iv) an expected answer. The first two parts are fixed for all tasks, while the question and answer are task-specific. In the fine-tuning stage, only the answer was trained in an autoregressive manner. The model is fine-tuned using the AdamW optimizer for 2 epochs [18], with the following hyper-parameters: $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate and batch size are varied with the size of the model. The learning rate is decayed with a cosine schedule and a warmup ratio of 0.03.

Benchmark OmniNA with task-specific deep learning methods

To assess OmniNA's question–answer (QA) performance, we conducted a comparative evaluation against state-of-the-art deep learning methods. The benchmark methods employed for specific tasks are detailed in Supplementary Table S6. For the detection of human and mice DNA elements, including promoters, enhancers, silencers, and insulators, we employed benchmark methods such as PromID [40], iEnhancer-DEEP [41], DeepSilencer [24], and CTCFBSDB [42]. Activation promoter and enhancer tasks were benchmarked with MPRA-DragoNN [28] and DeepSTARR [43]. PAS signal, TIS signal, and splice site detection tasks were benchmarked with DeepGSR [29] and SpliceRover [44]. CRISPR–Cas13d off-target effects were benchmarked with TIGER [8]. Chromatin accessibility, histone occupation, and TF binding prediction tasks were benchmarked with Enformer [7], Sei [45], and ExPecto [46]. Pathogenic SNP detection tasks were benchmarked with EVE [47]. Pathogenic virus classification and species detection tasks were benchmarked with the DeePaC model [34] and DeepMicrobes [35]. Input sequences were centered on the region of interest and resized by truncation or padding to meet each model's input length requirement.

A total of 10 metrics are used for the evaluation, including precision, recall, F1 score, accuracy, false positive rate, true positive rate, false negative rate, true negative rate, and Matthews correlation coefficient (MCC). In multilabel classification tasks such as histone, TF, and bacteria taxonomy classification, prediction accuracy for each label is individually assessed, treating any class with a different label as a negative sample.

Saturation mutation analysis

We quantify mutation effects using log-likelihood (LL) differences between mutated (s^{mut}) and wild-type (s^{WT}) nucleotide sequences:

$$\text{Predicted mutation effect} = \left| \frac{\left(\text{LL}\left(s^{\text{WT}}\right) - \text{LL}\left(s^{\text{mut}}\right)\right)}{\text{LL}\left(s^{\text{WT}}\right)} \right|,$$

where the LLs are estimated using the OmniNA-1.7B model, representing the predicted LL of the positive target in the respective fine-tuning task:

$$\text{LL(s)} = \sum\nolimits_j \log \Pr(x_j = S_j|S).$$

Benign and pathogenic mutations within the DNA sequences of PAS and TIS signal regions are sourced from the ClinVar database [32]. The actual CRISPR–Cas13d off-target effect ($S_{\text{off-target}}$) for each gRNA position is calculated as

$$S_{\text{off-target}} = 1 - S_{\text{gRNA}},$$

where gRNA relativity activity (S_{gRNA}) is collected from a previous study [8]. Domain regions of the BRCA1 and CHD7 proteins are obtained from the UniProt where gRNA database [8]. Full-length structures of BRCA1 and CHD7 spanning residues 200–1200 are predicted using AlphaFold2 [48].

Statistical analyses

We generally employed the t-test for the statistical analysis if unspecified. The paired t-test is used for assessing the differences between paired samples. Data distributions were determined through a permutation test involving 1000 permutations. For each permutation, the selected sample numbers were based on the data size. A significance threshold of P-value <0.05 is adopted for all tests, and these are two-sided in nature. We adjust P-values using the false discovery rate method to account for multiple comparisons.

Results

An overview of OmniNA

We developed OmniNA as a foundation model for nucleotide sequence (Fig. 1). The cognitive core of OmniNA is an LLM developed through incremental training on an extensive nucleotide sequence corpus, inheriting the benefits of emergent abilities of LLM and domain-specific knowledge. The backbone of OmniNA is a transformer-based autoregressive decoder with a self-attention mechanism [11]. We developed three models with varying layer and head numbers, spanning scales from 66 M to 330 M and up to 1.7 B parameters (Supplementary Fig. S2). The detailed network structure is delineated in Supplementary Fig. S3. We curated a dataset for model pre-training, which comprises 1076.2 B bases from 91.7 M nucleotide sequences across diverse species, extracted from the NCBI NT database (Fig. 1A; see the "Materials and

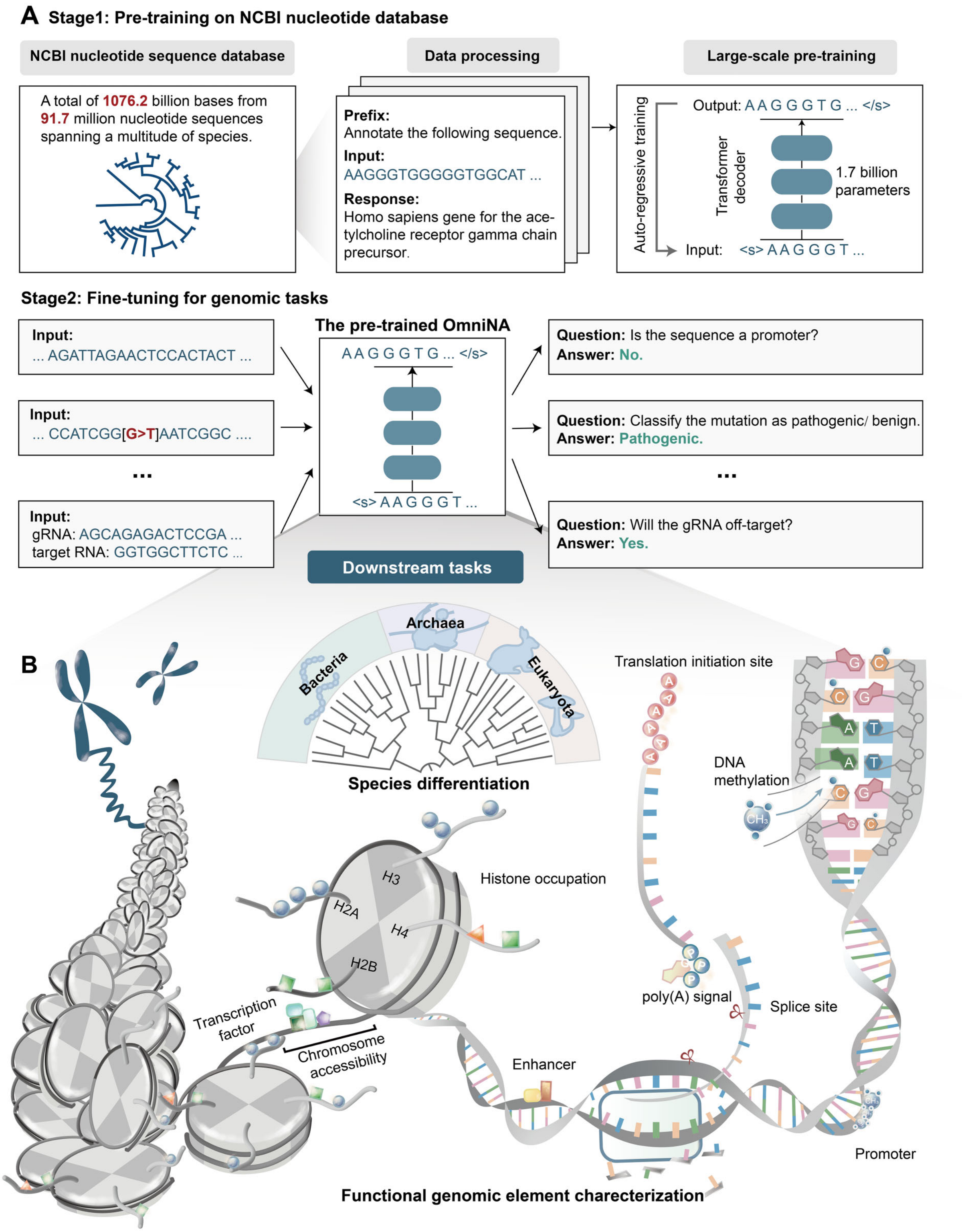


Figure 1. Schematic of OmniNA. (**A**) Nucleotide sequences, sourced from the NCBI NT database, are amalgamated with their corresponding text annotations for model pre-training. The model, featuring a transformer-based decoder, undergoes pre-training through an autoregressive approach. Fine-tuning is executed across 23 tasks in a QA framework. (**B**) Overview of the downstream tasks encompassing nucleotide sequence detection, ranging from taxonomy classification to genomic element detection.

methods" section). We designed prompts to structure the sequences and their corresponding natural language descriptions as model input (Fig. 1A and Supplementary Table S3). The input underwent chunking into subword tokens, followed by embedding through the token embedding layer. Token embeddings were then concatenated with position embeddings and fed into the transformer layers, followed by a regression layer for autoregressive training (Fig. 1A and Supplementary Fig. S3) [5]. The pre-trained OmniNA can handle versatile downstream applications in a multi-task manner (Fig. 1B). After fine-tuning, the model can manage taxonomy and genomic element identification tasks in a QA paradigm (Fig. 1A).

The generation performance of OmniNA

In this section, we systematically evaluate OmniNA's generation capabilities across four aspects, including the fidelity, diversity, reproducibility, and stability of the generated content. The fidelity evaluation focused on the model's ability to generate sequences and annotations consistent with ground truth counterparts. For sequences exhibiting high conservation (cosine similarity >0.65), OmniNA consistently produced highly similar sequences across different scales. For sequences exhibiting low conservation (cosine similarity <0.65), the largest model shows the most notable improvement in fidelity (Supplementary Fig. S5). To further evaluate fidelity in annotation generation, three annotation tasks were employed, including genomic type, species, and enzyme annotations. We observed an upward trend in generations' F1 score with the increasing model size (Fig. 2A and B), with OmniNA-1.7B achieving the highest F1 scores of 0.66, 0.24, and 0.59 for genomic type, species, and enzyme annotation tasks, respectively (Fig. 2A and Supplementary Fig. S5). Diving into diversity assessment, larger models demonstrated an increased capacity to generate diverse tokens, reflected in elevated token numbers and Shannon entropy (Fig. 2C and D). Particularly, larger models exhibited a heightened likelihood of generating low-frequency words (Fig. 2C and D). Our reproducibility analysis revealed that larger models exhibited higher consistency in generating annotations across different repeats, evident in increased cosine similarity (Fig. 2E). Visual inspection indicated that sentences generated by OmniNA-1.7B with similar semantics were more likely to cluster together, and annotations for the same sequence were closer (Fig. 2F). The variability in annotations for identical sequences largely arises from subtle distinctions. For example, the discrepancies can occur between different species within the same genus (Fig. 2F). Evaluation of generation's stability indicated that different truncations of the same sequence were more prone to yield similar annotations (Fig. 2G). For instance, sequences from the same genomes of HIV produced more similar annotations than those from different genomes (Fig. 2H). Notably, the model consistently generated correct annotations in both non-overlapping and overlapping regions of gag and pol genomes of HIV. In summary, our comprehensive evaluation reveals that the OmniNA model excels in consistently generating realistic and reasonable annotations.

Enhanced sequence and annotation representations by OmniNA

To gain a deeper understanding of OmniNA's capabilities in aligning sequence and textual patterns with natural language, we investigated the token representations of the pre-trained OmniNA-1.7B (see the "Materials and methods" section). The t-SNE plot reveals the model's ability to discriminate tokens representing distinct semantics, such as genus, genes, and sequences (Fig. 3A). Besides, tokens for gene symbols and gene functions exhibit distinct clustering, while sequence tokens with different base preferences manifest in separate loci (Fig. 3A). Moreover, taxonomy names are distinctly separated, with tokens from the same taxonomy clustering together (Fig. 3B). We also observed that viruses from different phyla form distinct subclusters (Fig. 3B).

Our further exploration of the feature representation space extended to three specific tasks in terms of sequence conservation delineation, species differentiation, and functional sequence differentiation. The ARI indexes underscore that different layers manifest optimal representation capabilities for distinct tasks (Fig. 3C). Specifically, the second layer can segregate sequences with varying conservation levels (Fig. 3D). The third layer can separate sequences from different superkingdoms (Fig. 3E). Leveraging bacteria, archaea 16s RNA (Fig. 3F), and fungi ITS sequences (Supplementary Fig. S6A), the second layer enables species discrimination at class level. Phylogenetic trees constructed based on the sequence representations further confirm the model's representation ability at both class and phylum levels (Fig. 3G and Supplementary Fig. S6B). The eighth layer distinguishes different branches of the evolutionary progress of SARS-CoV-2 (Fig. 3H). The phylogenetic trees underscore the model's ability to recover the evolutionary trajectory (Supplementary Fig. S6C). Quantitatively, the similarity with the first recognized strain (Wuhan-Hu-1/2019) significantly diminishes with the evolution of branches (Fig. 3I). The second and twenty-third layers can separate coding sequences of different functional proteins of viruses and bacteria, respectively (Fig. 3J and Supplementary Fig. S6D). Deeper layers are more effective at distinguishing complex DNA sequence features; specifically, t-SNE visualization demonstrates the twenty-third layer's capability in separating genomic elements based on their distinct functional roles (Supplementary Fig. S6E); the 20th layer is best for identifying CpG content (Supplementary Fig. S6F); the 23rd layer is most adept at differentiating non-B DNA structures (Supplementary Fig. S6G); and the 17th layer is prove for most effective performance in separating various RNA functional elements (Supplementary Fig. S6H).

We further benchmarked OmniNA-1.7B against current state-of-the-art nucleotide sequence foundation models, including DNABERT2 [20], Nucleotide Transformer-v2 [21], and Hyena-DNA [22], in terms of feature representation performance. For each model, ARI scores of the embedding and transformer layers across various sequence classification tasks are provided in Supplementary Table S7. Notably, OmniNA-1.7B outperforms all other models across 11 feature representation tasks (Supplementary Fig. S6I and J). Collectively, these findings highlight OmniNA-1.7B's capacity to assimilate biological knowledge, encompassing both textual patterns and nucleotide sequences.

We also observe empirical robustness when pretraining annotations are corrupted or sparse: taxonomic separability in the sequence embeddings is preserved (UMAP), with ARI decreasing from 0.18 (correct) to 0.12 (shuffled) and 0.09 (sparse) (Supplementary Fig. S7).

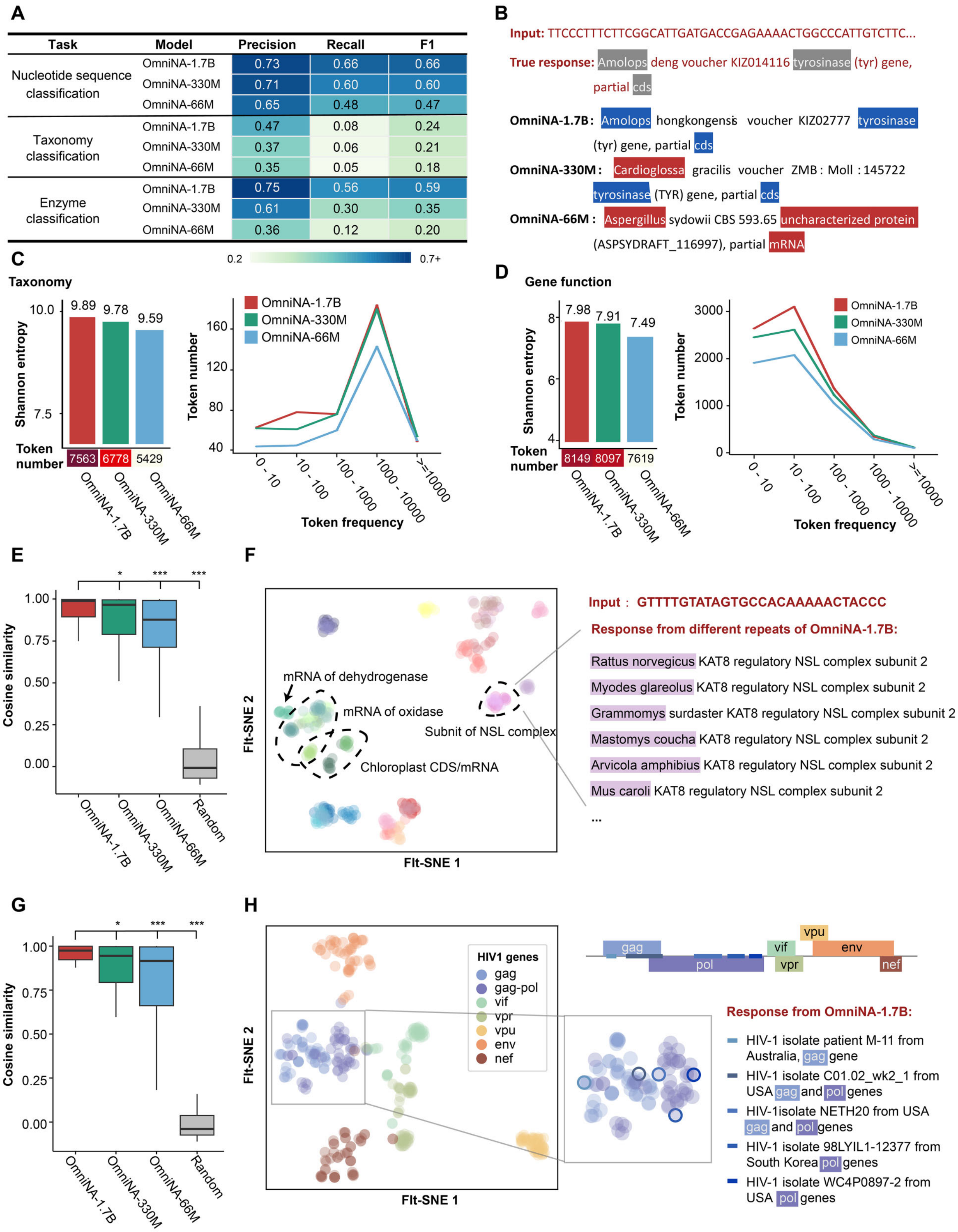


Figure 2. Evaluation of OmniNA's generation performance. (**A**) Fidelity assessment in sequence annotations generated by OmniNA models. For each target and model, the precision, recall, and F1 score represent the mean score across all labels. (**B**) Example of generated annotations for the same input by OmniNA models of varying sizes. (**C**) (Left) Bar plot illustrating Shannon entropy for taxonomy token generation. The heatmap at the bottom represents the generated token numbers. (Right) Line chart depicting generated token numbers correlated with token frequency. (**D**) (Left) Bar plot illustrating Shannon entropy for gene function token generation. The heatmap at the bottom represents the generated token numbers. (Right) Line chart depicting generated token numbers related to token frequency. (**E**) Box plot showcasing the cosine similarity of feature representations among different annotations generated from the same sentence. (**F**) t-SNE plot deciphering feature representations of generated annotations, with annotations from the same sequence colored identically. (**G**) Box plot reflecting the cosine similarity of token embeddings among different annotations generated from different truncations of the same sequence. (**H**) t-SNE plot deciphering the feature representations of generated annotations, with annotations from the same genome colored identically. $***P$-value <0.001; $**P$-value <0.01; $*P$-value <0.05.

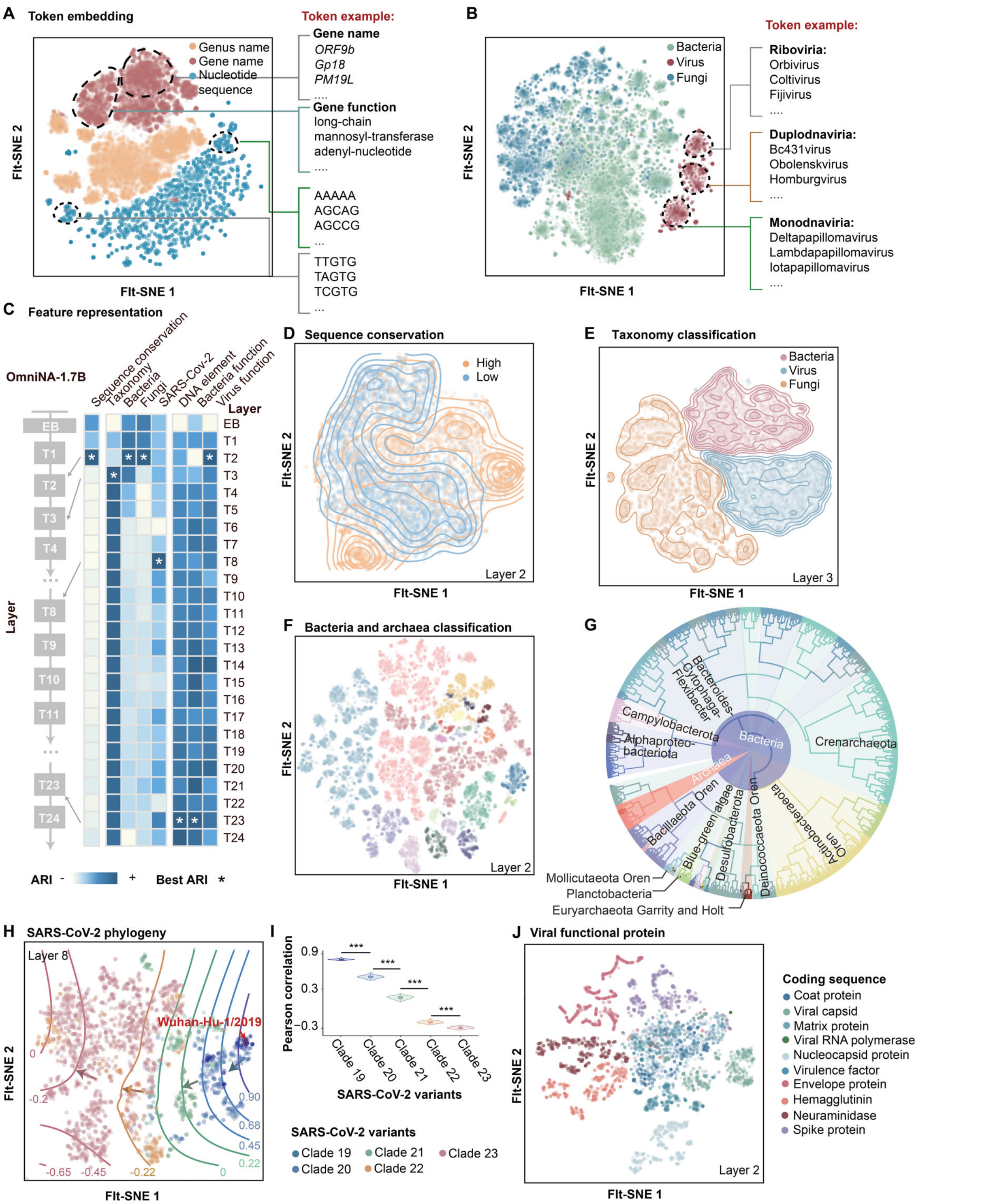


Figure 3. Sequence and token representations analysis. t-SNE plots visualize the word embeddings of tokens with distinct semantics (**A**) and superkingdoms (**B**). Each point represents a word, color-coded by its semantic group (**A**) and superkingdom (**B**). (**C**) Heatmap illustrates the ARI values for transformer layers across different representation tasks. The ARI scores are scaled by task, with asterisks denoting layers exhibiting the highest AIR for each task. (**D**) t-SNE plot visualizes sequences with varying sequence conservation represented by transformer layer 2. Each point represents a sequence, color-coded by sequence conservation. The counter plot reflects the density distribution of sequences of distinct conservation. (**E**) t-SNE plot visualizes sequences from different superkingdoms represented by transformer layer 3. The counter plot depicting the density distribution of sequences from different superkingdom categorizations. (**F**) t-SNE plot visualizes bacterial sequences represented by transformer layer 2. The color-coded according to bacterial class. (**G**) Phylogenetic tree of the bacteria constructed with the similarity between sequence representations. Line segment colors denote bacterial class. The outer pie chart colors signify bacterial phylum, and inner pie chart colors distinguish between bacteria and archaea. (**H**) t-SNE plot visualizes sequences from different SARS-CoV-2 variants represented by transformer layer 8. Counter lines reflect the Pearson correlation of each strain with the first recognized strain (Wuhan-Hu-1/2019), with the red-highlighted point representing Wuhan-Hu-1/2019. (**I**) Violin and box plot illustrate the Pearson correlation between different SARS-CoV-2 clades and the first recognized strain. Data is randomly sampled 1000 times from each branch, with 100 strains per sample, and the average Pearson correlation is shown. The boxplot presents the median, upper, and lower quartiles of the distribution. (**J**) t-SNE plot visualizes virus sequences represented by transformer layer 2. $***P$-value <0.001.

Comprehensive evaluation of OmniNA's generalization across nucleotide sequence tasks

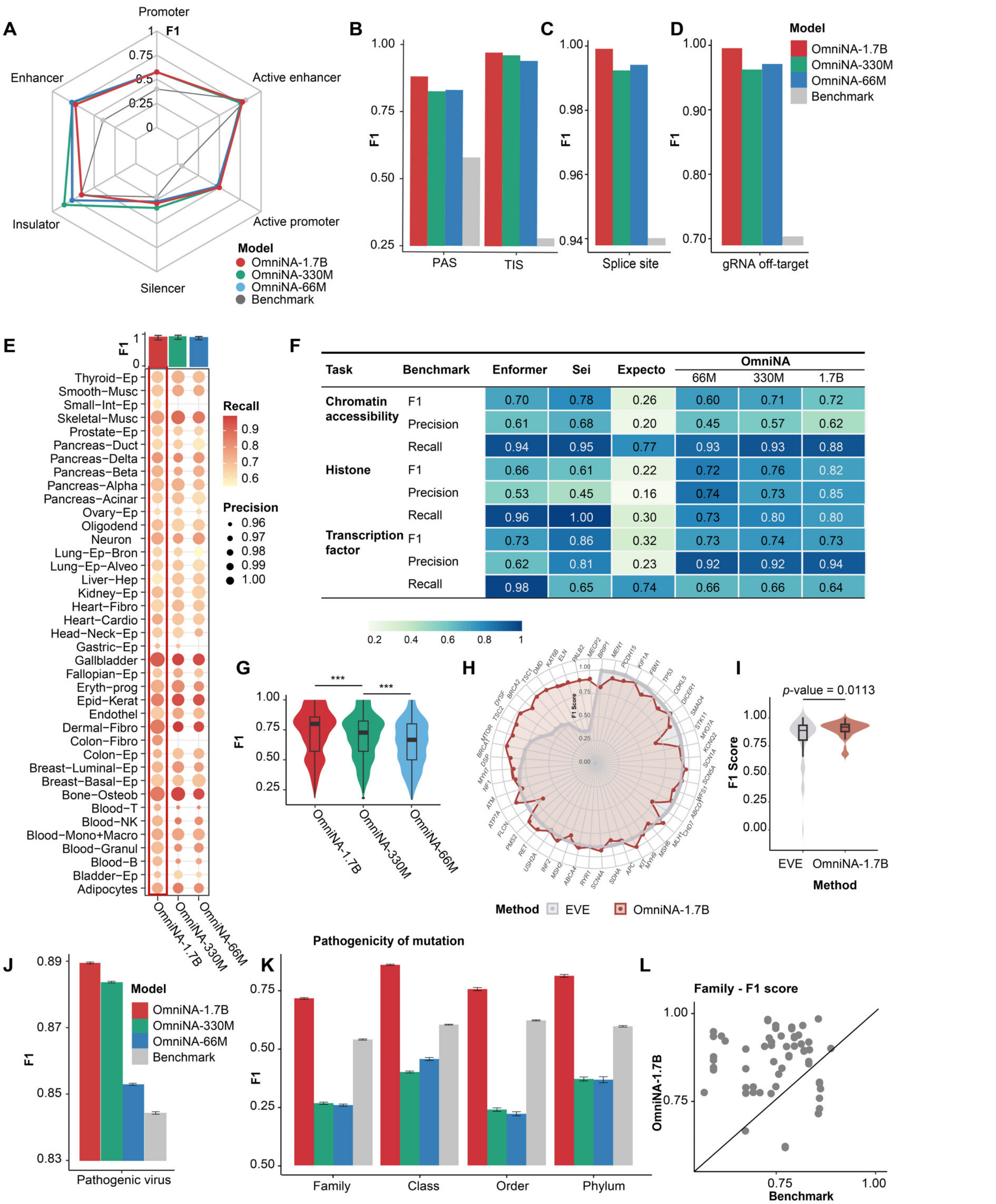
To assess the generalization capabilities of OmniNA on nucleotide sequence tasks, we conducted fine-tuning experiments on 23 diverse tasks encompassing genome and species detection. All downstream fine-tuning and evaluations reported here use sequence-only inputs, with no annotations supplied during training or testing; hence, the reported accuracy reflects annotation-free deployment. OmniNA generally achieved better F1 scores than benchmark methods (Fig. 4). Specifically, for tasks related to human DNA element identification, OmniNA achieved comparable or superior performance as compared with the benchmark methods (Fig. 4A). Besides, OmniNA-1.7B excelled in PAS, TIS signal, and splice site detection, achieving remarkable F1 scores of 0.88, 0.97, and 0.98, respectively (Fig. 4B and C). The superior performance of the OmniNA-1.7B model can also be observed for gRNA off-target detection with an F1 score of 0.98 (Fig. 4D and Supplementary Fig. S8A). In methylated region detection tasks, the OmniNA-1.7B model achieved comparable F1 and higher precision scores across various cell types compared with the 66 M and 330 M models (Fig. 4E). Additionally, for chromatin feature prediction, OmniNA-1.7B delivered high-accuracy predictions for chromatin accessibility and histone modification, surpassing current state-of-the-art methods (Fig. 4F, Supplementary Fig. S8B and C, and Supplementary Table S8). Our method further demonstrated high performance in the prediction for TF occupation, achieving a median F1 score of 0.73 across all TF targets (Fig. 4F, Supplementary Fig. S8D, and Supplementary Table S8). Besides, the model's performance on pathogenic variation detection tasks improved with increasing model size (Fig. 4G). We compared the OmniNA-1.7B model against the EVE method. Across 48 genes, OmniNA-1.7B achieved higher F1 scores in 27 genes compared with EVE (Fig. 4H). Overall, OmniNA-1.7B achieved a significantly higher mean F1 score of 0.90 compared with 0.81 for EVE (Fig. 4I; two-sided *t*-test, *P*-value = 0.011). OmniNA also exhibited proficiency in species detection tasks (Fig. 4J–L). In the pathogenic virus detection task (Fig. 4J), the F1 scores of OmniNA-1.7B surpassed those of benchmark methods. Furthermore, on bacteria taxonomy classification tasks related to class, family, order, and phylum (Fig. 4K and L and Supplementary Fig. S8E), OmniNA-1.7B outperformed the benchmark model. We also evaluated the model on non-B DNA structure classification, where OmniNA-1.7B achieved an Accuracy@1 of 0.78 and Accuracy@3 of 0.88 (Supplementary Fig. S9A). Furthermore, we extended our evaluation to microbiology tasks. For bacterial functional-sequence sub-classification, including ARGs and VGs, we report top-k performance across five ARG mechanisms and 11 VG types. OmniNA-1.7B achieved an Accuracy@1 = 0.59 and F1@1 = 0.56, which improved to Accuracy@3 = 0.92 / F1@3 = 0.54 for ARG mechanisms (Supplementary Fig. S9B and Supplementary Table S9). For VG classification, the model reached Accuracy@1 = 0.93 and F1@1 = 0.87 and further improved to Accuracy@3 = 0.96 / F1@3 = 0.63 (Supplementary Fig. S9C and Supplementary Table S9). For host specificity prediction, OmniNA-1.7B accurately classified both bacteriophage and eukaryotic virus hosts. In bacteriophage host prediction (12-way), the model achieved Accuracy@1 = 0.73 and F1@1 = 0.78, improving to Accuracy@3 = 0.97 and F1@3 = 0.60 under top-3 evaluation (Supplementary Fig. S9D and Supplementary Table S9). In eukaryotic virus host prediction (15-way), it achieved Accuracy@1 = 0.78 and F1@1 = 0.69, with Accuracy@3 = 0.91 and F1@3 = 0.51 (Supplementary Fig. S9E and Supplementary Table S9). On cfRNA pancreas tumor vs. normal distinguishing task, the model achieved near-perfect discrimination (Accuracy/F1/Precision/Recall/MCC ≈1.0), and its learned embeddings showed clear, complete separation between tumor and normal samples in UMAP space (Supplementary Fig. S9F). All detailed quantitative benchmark results are listed in Supplementary Table S10.

Discriminating mutation effects in distinct genomic contexts with OmniNA

After fine-tuning OmniNA-1.7B on diverse nucleotide learning tasks aligned with natural language paradigms, we further verified the model's ability to predict the mutation impact within varied genomic contexts. The effect score of a mutation was defined as the OmniNA-1.7B estimated LL between mutated and wild-type nucleotide sequences (Fig. 5A). Our initial focus was on mutations influencing PAS and TIS signals. Analyzing mutations in genomic loci around PAS regions reveals a consistent increase in the odds ratio (OR) between pathogenic and benign mutations as predicted mutation effect increases (Fig. 5B; Pearson correlation = 0.92, *P*-value = 9e−3). Similarly, mutations at TIS-associated genomic loci exhibit a proportional rise in the OR between pathogenic and benign mutations with escalating predicted mutation effects (Fig. 5C; Pearson correlation = 0.92, *P*-value = 1e−2). Furthermore, OmniNA-1.7B captures the mutation effects at the 23 positions of CRISPR–Cas13 gRNA, showcasing significant alignment with actual mutation effects (Pearson correlation = 0.65, *P*-value = 6.5e−4). Moreover, OmniNA-1.7B showcases discriminative power in identifying pathogenic effects at functional regions within protein coding sequences. For instance, the mutation effect in the domain region of the *BRCA1* coding sequence significantly surpasses that in non-domain regions (Fig. 5D; *P*-value <1e−5). Structurally, the pathogenic effects across three domains outperform those in surrounding ordered and disordered structures (Fig. 5E and F). Furthermore, we extend this discriminative capability to *CHD7*. The mutation effect in the domain region of CHD7 exhibits a significant elevation compared with non-domain regions (Fig. 5G; *P*-value <1e−5). This trend is consistent with the structural analysis, where the pathogenic effects across the three domains of CHD7 surpass those found in surrounding regions (Fig. 5H and I). Crucially, OmniNA-1.7B can also discern the functional impact of synonymous mutations. When fine-tuned on a dataset of annotated pathogenic and benign synonymous variants across 24 genes, the model achieved a mean F1 score of 0.68, demonstrating its capacity to capture sequence context effects beyond amino acid changes (Supplementary Fig. S10). In conclusion, OmniNA-1.7B exhibits a grasp of the structural and functional intricacies of nucleotide sequences, enabling precise predictions regarding how mutations alter fundamental aspects, showcasing its potential in mutation effect prediction.

Ablation study for few-shot learning and cross-species generalization analysis

We conducted an extensive evaluation of the OmniNA's few-shot learning capabilities. Training with the histone



Task	Benchmark	Enformer	Sei	Expecto	OmniNA 66M	OmniNA 330M	OmniNA 1.7B
Chromatin accessibility	F1	0.70	0.78	0.26	0.60	0.71	0.72
	Precision	0.61	0.68	0.20	0.45	0.57	0.62
	Recall	0.94	0.95	0.77	0.93	0.93	0.88
Histone	F1	0.66	0.61	0.22	0.72	0.76	0.82
	Precision	0.53	0.45	0.16	0.74	0.73	0.85
	Recall	0.96	1.00	0.30	0.73	0.80	0.80
Transcription factor	F1	0.73	0.86	0.32	0.73	0.74	0.73
	Precision	0.62	0.81	0.23	0.92	0.92	0.94
	Recall	0.98	0.65	0.74	0.66	0.66	0.64

Figure 4. Performance of fine-tuned OmniNA on nucleotide sequence identification tasks. (**A**) Radar plot shows the F1 score achieved by OmniNA and the benchmark method on the task of DNA element detection. Bar plot illustrating the F1 score achieved by OmniNA model and the benchmark method on the task of splice site detection (**B**), PAS/TIS site detection (**C**), and gRNA off-target prediction (**D**). (**E**) Dot plot depicts the performance of OmniNA with different parameters on the task of methylated region detection. The circle size reflects the precision score of method, and the color reflects the recall of the method. The bar plot in the above panel shows the F1 score of the OmniNA model with different parameters. (**F**) The performance of OmniNA and three benchmark methods, i.e. Enformer, Sei, and ExPecto, on chromatin feature detection tasks. (**G**) The boxplot presents the median, upper, and lower quartiles of the F1 score for the pathogenic variation detection performance across all genes. (**H**) Radar plot shows the F1 score of the OmniNA-1.7B model and benchmark method for pathogenic SNV detection. (**I**) Violin and box plots comparing the overall F1 score distributions between EVE and OmniNA-1.7B. *T*-test was applied for the analysis. (**J**) Bar plot illustrating the F1 score achieved by OmniNA and the benchmark method on the task of pathogenic virus detection. Data are randomly sampled 1000 times from each branch, with 1000 strains per sample. The center line represents the mean, and the error bar indicates the 95% CI. (**K**) Bar plot illustrating the F1 score achieved by OmniNA and the benchmark method on the task of bacteria species classification. Data are randomly sampled 1000 times from each branch, with 1000 strains per sample. The center line represents the mean, and the error bar indicates the 95% CI. (**L**) The F1 score comparison between the OmniNA-1.7B model and benchmark method on bacteria phylum classification task. ∗∗∗*P*-value <0.001.

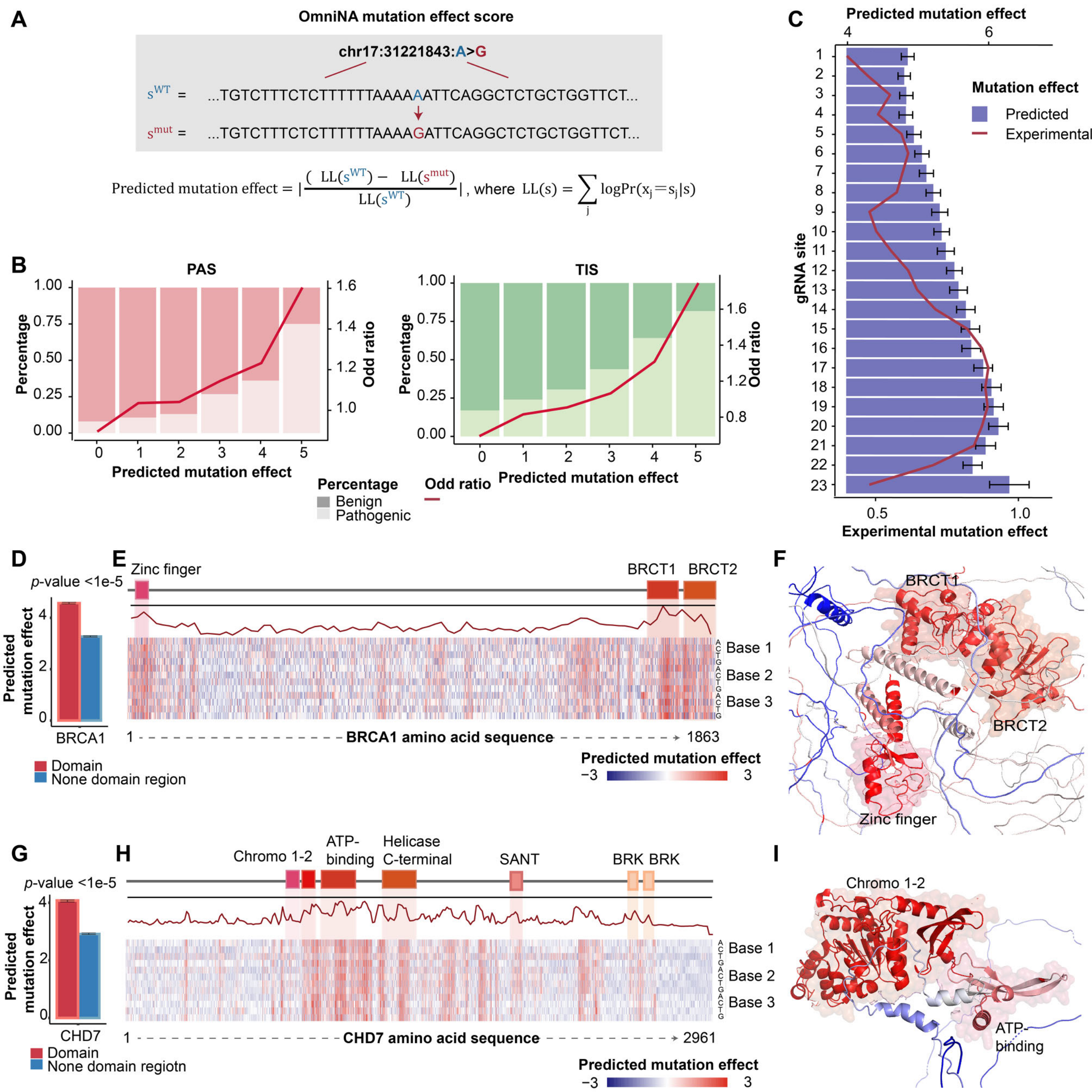


Figure 5. Predicted mutation effects of clinically and functionally relevant missense variants. (**A**) Schematic diagram of the mutation effect score calculation. (**B**) Bar plot illustrates the correlation between mutation effect scores and the proportion of pathogenic and non-pathogenic mutations. Line plot depicts the OR between pathogenic and benign mutations. (**C**) Bar plot shows the mean predicted mutation effects for each gRNA position, with error bars representing the 95% confidence interval (CI) of the mutation score. (**D**) Bar plot shows the mean mutation effect score for mutations in the domain region compared with the non-domain region of *BRCA1* coding gene. The error bar represents the 95% CI of the distributions of mutation scores. (**E**) Upper panel, schematic representation of the domains and non-domain protein coding regions of *BRCA1*; middle panel, maximum mutation effect score observed for each nucleotide base; lower panel, heatmap illustrating mutation effect scores across all possible 3 × 4 missense variants. (**F**) AlphaFold prediction of the three-dimensional structure of the BRCA1 protein. The residues are colored by the predicted mutation effect scores. (**G**) Bar plot shows the mean mutation effect score for mutations in the domain region compared with the non-domain region of *CHD7* coding gene. The error bar represents the 95% CI of the distributions of mutation scores. (**H**) Upper panel, schematic representation of the domains and non-domain protein coding regions of *CHD7*; middle panel, maximum mutation effect score observed for each nucleotide base; lower panel, heatmap illustrating mutation effect scores across all possible 3 × 4 missense variants. (**I**) AlphaFold prediction of the three-dimensional structure of the CHD7 protein. The residues are colored by the predicted mutation effect scores.

occupation profiles, we found that larger model consistently outperformed its counterparts as the percentage of training data increased (Supplementary Fig. S11A). Notably, the OmniNA-1.7B model demonstrated comparable performance with 75% and 100% of the data, showcasing its effectiveness in learning histone occupation profiles from limited information. To further assess OmniNA's cross-species generalization capability, we fine-tuned OmniNA on human DNA element classification tasks and applied it for mouse DNA element classification. As illustrated in Supplementary Fig. S11B, OmniNA achieved comparable performance across all evaluations with the benchmark methods trained specifically with mice sequences. Quantitative results are detailed in Supplementary Table S11. These findings underscore OmniNA's robust few-shot learning and impressive generalization capacities.

Discussion

Deep learning is currently undergoing a paradigm shift toward foundation models, capable of generalizing across a wide range of downstream tasks and yielding emergent capabilities. In the domain of nucleotide sequencing, understanding the intricate rules governing sequences is crucial, and there is a growing need to align various nucleotide learning tasks with natural language. In this context, the development of our foundation model OmniNA not only addresses the technical challenges posed by the data deluge but also tackles the inherent limitations of traditional bioinformatic methodologies. By deploying a foundation generative model, our objective is to provide biologists and researchers with a versatile tool that can autonomously discern intricate relationships within nucleic acid sequences. This model not only facilitates the effective capture of inter-sequence correlations but also establishes a robust foundation for subsequent downstream tasks.

We observed a positive correlation between the model size and OmniNA's generation capabilities. As model size increased, we noted a corresponding improvement in the fidelity, diversity, reproducibility, and stability of generated annotations. Notably, the diversity of low-frequency words increased with model size, emphasizing the capacity of larger models to effectively handle a broader range of vocabulary (Fig. 2C and D). The reproducibility was further highlighted by the consistent sequence annotations generated across multiple repetitions. We further demonstrated the model can preserve the semantic similarity between different sequence annotations (Fig. 2F). Regarding the stability of the generations, we proved that various truncations of a single sequence consistently produced similar representations (Fig. 2H). The generated annotations exhibited a degree of overlap for genomes with overlapping regions, highlighting the model's ability to capture the intricacies of shared genomic elements (Fig. 2H). In summary, larger model sizes positively correlate with OmniNA's improved generation capabilities, particularly in their ability to handle a broader vocabulary and capture the intricacies of genomic elements.

Our study provides evidence that OmniNA learns representations of tokens and sequences across diverse biological contexts. In the absence of supervision, our model exhibits a capacity to reveal nuanced sensitivity to different biological structures and evolutionary information. As transformer layers deepen, the model's ability to represent complex biological information becomes more evident. Shallow layers appear proficient in capturing elementary information, such as sequence conservation and species distinctions. The intermediate layers demonstrate a heightened capacity to discern the evolutionary dynamics of the SARS-CoV-2. The deeper layers capture information related to sequence functionality. Compared with sequence-only foundation models, such as DNABERT2 [20], Nucleotide Transformer-v2 [21], and Hyena-DNA [22], OmniNA-1.7B demonstrates superior feature representation capabilities, particularly in tasks requiring fine-grained sequence discrimination. This is particularly evident in its ability to discern the evolutionary dynamics of SARS-CoV-2 and DNA elements. The observed performance gains may stem from OmniNA's unique design, which integrates both sequence and annotation-based information. This dual modality allows for complementary information exchange between sequences and annotations, enabling the model to develop a more holistic understanding and representation of biological data.

In contrast to canonical deep learning approaches in genomics that learn from specific tasks, OmniNA is a general solution applicable to a wide range of tasks, including species classification and DNA and RNA processing. OmniNA is a versatile model capable of handling various tasks without the need for task-specific architectures. The multitask OmniNA model reduces the complexity and effort required in designing and training specialized models. The flexibility of large-language models in accommodating diverse tasks in a QA manner enables a broader range of users, including those with less specialized expertise in deep learning, to leverage the power of advanced language models for diverse applications [49]. Furthermore, OmniNA is often optimized for high precision, ensuring that its positive predictions are highly reliable. While this may result in lower recall in some tasks, this characteristic makes it an ideal tool for high-confidence screening, where minimizing false positives is critical to guide expensive and time-consuming experimental validation. In this role, OmniNA can complement other methods, serving as a robust first-pass filter to prioritize candidates for deeper investigation with specialized tools. Besides, the integration of diverse species and genomic annotations into the training data has significantly enhanced OmniNA's ability to generalize across a wide array of downstream tasks. In our evaluation of fine-tuning OmniNA on 23 tasks, the model demonstrated comparable or superior performance to dedicated deep learning models in specific tasks, highlighting its adaptability in genomic contexts. Importantly, our findings suggest that different-scale models achieve similar results in straightforward tasks, such as DNA element detection. The superior performance of larger-scale models becomes evident in more challenging tasks like pathogenic mutation identification. This scalability feature positions OmniNA as a powerful tool, demonstrating its ability to learn intricate nucleotide sequence grammar and translate the knowledge into mutation effect prediction. This emphasis on high precision indicates that OmniNA could serve as a useful preliminary screening tool for prioritizing high-confidence candidates for experimental validation in contexts where minimizing false positives is important. This capability enhances its utility in understanding the functional implications of genetic variations. In summary, OmniNA emerges as a versatile and scalable solution, empowering users across various expertise levels to navigate genomics with ease and efficacy.

This study has several limitations. First, the scale of deep learning models exhibits a power law correlation with performance, connecting model and data sizes to model performance [50]. As genomics research advances, the integration of OmniNA with expanding model sizes into the dynamic realm of high-throughput sequencing technologies requires amplifying both model and data scales. Second, fine-tuning performance can be influenced by the quality and completeness of annotations used during pretraining. Although OmniNA leverages annotations only as weak cues and performs sequence-only inference, noisy labels can still affect learned representations. Our control analysis (Supplementary Fig. S7) shows that embeddings remain taxonomically separable under correct, shuffled, or sparse annotations, indicating resilience but non-negligible degradation. Future work will implement self-correction through versioned corpus updates, quality-weighted training, and drift checks on curated sequence-only benchmarks. Third, when comparing OmniNA with existing models, potential overlap between OmniNA test sequences and the undisclosed training sets of certain baselines cannot be ruled out. OmniNA itself uses strictly non-overlapping splits, but the lack of standardized or publicly documented partitions for several baselines (e.g. Enformer, Sei, and ExPecto) makes cross-model leakage unverifiable. Any such overlap would inflate baseline rather than OmniNA performance, suggesting that our comparative advantage is conservative. Additionally, for regression models, including MPRA-DragoNN, Enformer, and ExPecto, optimal AUC values from training data were employed to determine cutoffs for converting tasks into classification tasks. These factors could introduce evaluation biases. Besides, the inclusion of domain-specific annotations in the pre-training phase could introduce potential bias, which might affect the model's generalization ability. To mitigate this, we ensured that downstream evaluations were conducted using sequence-only inputs and demonstrated that OmniNA generalizes to tasks absent from the pre-training corpus. Lastly, OmniNA is currently limited to processing sequences of $\sim$3000 base pairs, which may restrict its ability to capture long-range regulatory interactions such as distal enhancer–promoter coordination and 3D chromatin effects. However, prior works (e.g. DeepSEA, 1 kb; Sei, 4 kb) indicate that 1–4 kb windows capture the majority of proximal regulatory information, including promoter motifs, splice sites, and local chromatin features, and OmniNA achieves state-of-the-art performance on local regulatory tasks despite the absence of long-range context. For applications requiring long-range modeling, this truncation represents a quantifiable limitation. Future iterations will incorporate extended receptive fields to improve modeling of long-range dependencies and broaden its applicability across diverse genomic contexts.

Conclusions

In conclusion, OmniNA represents an endeavor to leverage foundation models for comprehensive nucleotide learning across diverse species and genome contexts. Unlike existing models that focus primarily on sequence recognition, OmniNA integrates nucleotide sequences with their corresponding annotations, allowing the model to benefit from the complementary information provided by both. This integration enhances our understanding of nucleotide sequence rules and provides a versatile and user-friendly platform for addressing challenges in genomics and molecular biology. As we open-source it for collaborative exploration, we anticipate that OmniNA will pave the way for uncovering universal principles governing nucleotide sequences, offering novel insights and methodologies for exploring the intricacies of genomics.

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Author contributions: X.L. designed and supervised the study. X.L., X.S. and M.Y. wrote the manuscript. X.L., K.C. and L.S. revised the manuscript. X.L., J.L. and X.S. collected the data. X.S. and J.L. processed the data. X.L. and X.S. interpreted the results. All authors reviewed and approved the submission of this manuscript.

Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

None declared.

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Data availability

The source code of OmniNA is available at https://github.com/xilinshen/OmniNA and has been archived at Zenodo (DOI: 10.1101/2024.01.14.575543). The data for pre-training are openly available at https://ftp.ncbi.nlm.nih.gov/blast/db/FASTA/. The datasets supporting the conclusions of this article are included within the article and its additional files.

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